

MOHD FAISAL HUSSAIN

CONTACT

- 📍 Bhopal, India 462023
- ☎ +91-7819840241
- ✉ faisalhussain17122003@gmail.com

SKILLS

- Machine learning & deep learning: PyTorch, transformers, RNNs, LSTMs
- LLMs: Qwen, Ollama, prompt engineering, memory systems
- AI Infrastructure: FastAPI, SQLite, VectorDBs, Graph Structures
- NLP: summarization, classification, information extraction
- Systems optimization: CPU-bound inference, quantization
- Tools: Python, Git, Linux, model deployment

CERTIFICATIONS

- Andrew Ng's Deep Learning Specialization
- CS50x
- CS50 Python
- CS50 AI
- Jovian Deep Learning

LINKS

- [LinkedIn](#)
- [GitHub](#)

EDUCATION

01/2026
2023 - Current
B.Tech in Electronics & Communication
IIIT Bhopal, India
GPA: CPI: 7.63

AI/ML engineer specializing in applied AI infrastructure and graph algorithms. Expertise in resource-efficient LLM systems, context engineering, and iterative ML design. Proven ability to architect full-stack ML workflows—from edge data ingestion and LLM extraction to vector retrieval pipelines—that perform reliably in production.

PROJECTS

MindCache – Full-Stack AI Knowledge Graph System

Tech: Python, FastAPI, Ollama, PyTorch, NLP, Graph Algorithms, SQLite

- Engineered an end-to-end AI platform that passively ingests unstructured context (browsing, code, logs) and autonomously structures it into a queryable, multi-rooted knowledge graph.
- **Intelligent Preprocessing:** Built a custom tri-state Input Denoiser featuring a regex-based code skeletonizer and a fuzzy-hash log deduplicator, reducing LLM token context usage by up to 60%.
- **Graph Algorithms at Scale:** Developed a 1500+ line graph restructuring engine implementing root orthogonality, merge cycle resolution, token-aware bin-packing, and automated splitting of overloaded topic leaves.
- **Hybrid Retrieval Search:** Implemented a 5-phase generative search pipeline combining vector embeddings with a custom BM25 scorer (60/40 weighted), achieving **92% retrieval accuracy (47/51)** on complex long-term memory evaluations.
- **Agentic Reasoning:** Created a Decision Analyzer agent that cross-references temporal memories against episodic/knowledge context to autonomously track the state validity (active vs superseded) of past user decisions.
- **Data Engineering for Fine-Tuning:** Developed a data curation loop that transforms real-world system usage into a supervised fine-tuning dataset, enabling the specialization of small, resource-efficient local models.

Product Comparator – NLP-Driven Review Analysis

Tech: Python, NLP, Transformers, Sentiment Analysis

- Engineered an **aspect-based sentiment analysis (ABSA)** pipeline using Transformers to distill thousands of noisy, user-generated sentences into aggregated product comparisons.
- Implemented a custom **evidence-aggregation algorithm** that calculates confidence scores by weighting sentiment strength against review usefulness.
- Resolved real-world data issues by utilizing semantic clustering to reconcile conflicting opinions and handle sparse evidence for specific product aspects.
- Focused on **decision-support outputs**, not just sentiment labels, enabling clearer, robust product comparisons.